

# Qizhen Lan

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Houston, TX, United States

## RESEARCH INTERESTS

- My research develops **efficient, trustworthy, and agentic AI systems** at the intersection of medical imaging and computer vision. I work on 3D segmentation and medical foundation models, model compression and adaptation via knowledge distillation and pruning, and multimodal agents that support clinical reasoning and real-world workflows.

## EDUCATION

- **University of Alabama at Birmingham** Jan 2024 – May 2025  
*Ph.D. in Computer Science* Birmingham, United States
- **Bowling Green State University** Aug 2019 – Dec 2023  
*Ph.D. student in Data Science* Bowling Green, United States
- **Bowling Green State University** Aug 2018 – Jul 2019  
*Master of Science in Analytics* Bowling Green, United States
- **Bowling Green State University** Aug 2014 – Jun 2018  
*Bachelor of Science in Business Analytics and Intelligence* Bowling Green, United States

## PROFESSIONAL EXPERIENCE

- **University of Texas Health Science Center at Houston**  May 2025 – present  
*Postdoctoral Research Fellow (mentor: Dr. Xiaqian Jiang)* Houston, TX, United States
  - Pretrained **3D brain foundation models** on large-scale MRI cohorts for Alzheimer's-related downstream tasks.
  - Built a **fast whole-brain segmentation pipeline** with approximately **~1s inference**, reducing processing time from FreeSurfer-scale hours to deployment-level latency.
  - Designed region- and context-aware distillation methods for compact 3D segmentation models, achieving **sub-10M-parameter** students while preserving downstream performance.
  - **Outcomes:** One **npj Health Systems** paper accepted and four **MICCAI 2026** papers accepted, including two early accepts (top 9% among 4,601 submissions).
- **University of Alabama at Birmingham (UAB)**  Jan 2024 – May 2025  
*Research Assistant (Advisor: Dr. Qing Tian)* Birmingham, AL, United States
  - Led optimization and deployment of object detection models on NVIDIA embedded GPUs (e.g., Jetson Xavier and Orin) for real-time autonomous-driving perception.
  - Developed adaptive **knowledge distillation** methods for dense prediction and DETR-style detectors, reducing model parameters by up to **70%** while maintaining detection performance.
  - **Outcomes:** First-author papers in **ICCV 2025**, **WACV 2026**, and **WACV 2024**, with additional publications in WACV, ICASSP, IV, and T-IV.
- **Bowling Green State University (BGSU)**  Aug 2021 – Jan 2024  
*Research Assistant (Advisor: Dr. Qing Tian)* Bowling Green, OH, United States
  - Developed **efficient AI** algorithms for object detection models under autonomous-driving deployment constraints.
  - Built multi-factor and selective dark-knowledge distillation frameworks that reduce model size and computation while preserving detection accuracy across modern detectors.
  - **Outcomes:** IEEE T-IV journal paper and ICPR/WACV publications in efficient object detection and autonomous-driving perception.

- [M.1] **Qizhen Lan**, Yu-Chun Hsu, Nida Saddaf Khan, Xiaoqian Jiang. *Toward Clinically Deployable 3D Auto-Contouring: A Region- and Context-Aware Knowledge Distillation Framework*. **npj Digital Medicine**, 2026 (In preparation).
- [J.1] **Qizhen Lan**, Aaron Choi, Jun Ma, Bo Wang, Zhongming Zhao, Xiaoqian Jiang, Yu-Chun Hsu. *From Performance to Practice: Knowledge-Distilled Segmentator for On-Premises Clinical Workflows*. **npj Health Systems**, 2026.
- [J.2] Jung Im Choi\*, **Qizhen Lan**\*, Qing Tian. *Boosting Deep Detector Efficiency and Robustness through Detection Discriminant Reorganization and Compression*. **Neural Networks**, 2026 (\* Co-first author; Accepted).
- [J.3] **Qizhen Lan**, Qing Tian. *Instance, Scale, and Teacher Adaptive Knowledge Distillation for Visual Detection in Autonomous Driving*. **IEEE Transactions on Intelligent Vehicles (T-IV)**, 2022.
- [C.1] Zhisheng Chen, Yingwei Zhang, **Qizhen Lan**, Tianyu Liu, Huacan Wang, Yi Ding, Ziyu Jia, Ronghao Chen, Kun Wang, and Xinliang Zhou. *Uni-NTFM: A Unified Foundation Model for EEG Signal Representation Learning*. **ICLR**, 2026.
- [C.2] Jinwei Su\*, **Qizhen Lan**\*, Yinghui Xia, Lifan Sun, Weiyou Tian, Tianyu Shi, Xinyuan Song, and Lewei He. *Difficulty-Aware Agentic Orchestration for Query-Specific Multi-Agent Workflows*. **WWW**, 2026 (\* Co-first author).
- [C.3] Tingyu Wu, Zhisheng Chen, Ziyang Weng, Shuhe Wang, Chenglong Li, Shuo Zhang, Sen Hu, Silin Wu, **Qizhen Lan**<sup>†</sup>, Huacan Wang, and Ronghao Chen. *KnowMe-Bench: Benchmarking Person Understanding for Lifelong Digital Companions*. **ACL Main Conference**, 2026 (<sup>†</sup> Corresponding author; Oral Presentation).
- [C.4] Xi Xiao, Zhuxuanzi Wang, Chen Liu, **Qizhen Lan**, Lin Zhao, Jianyang Gu, Muchao Ye, Tianyang Wang, and Hao Xu. *CVD-DPO: Anchoring Grounded Reasoning via Counterfactual Visual Decoupling*. **ECCV**, 2026.
- [C.5] Mengchen Fan, Baocheng Geng, Xi Xiao, Tianyang Wang, Siyuan Mei, Pulin Che, Xiaoqian Jiang<sup>†</sup>, and **Qizhen Lan**<sup>†</sup>. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **MICCAI**, 2026 (<sup>†</sup> Corresponding authors; Early Accept).
- [C.6] Siyuan Mei, Yanteng Zhang, Yan Xia, **Qizhen Lan**, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Xiaoqian Jiang, Fuxin Fan, Yixing Huang, and Andreas Maier. *Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation*. **MICCAI**, 2026 (Early Accept).
- [C.7] **Qizhen Lan**, Qing Tian. *CLOCKDistill: Consistent Location-and-Context-aware Knowledge Distillation for DETRs*. **WACV**, 2026.
- [C.8] **Qizhen Lan**, Jong-Im Choi, Qing Tian. *Visual Detector Compression via Location-Aware Discriminant Analysis*. **WACV**, 2026.
- [C.9] Chen Liu\*, **Qizhen Lan**\*, Zhicheng Ding, Xinyu Chu, Qing Tian. *Learnable Instance Attention Filtering for Adaptive Detector Distillation*. **ICASSP**, 2026 (\* Co-first author).
- [C.10] **Qizhen Lan**, Qing Tian. *ACAM-KD: Adaptive and Cooperative Attention Masking for Knowledge Distillation*. **ICCV**, 2025.
- [C.11] Lijing Zhu, **Qizhen Lan**, Qing Tian, Wenbo Sun, Li Yang, Lu Xia, Yixin Xie, Xi Xiao, Tiehang Duan, Cui Tao, Shuteng Niu. *ETT-CKGE: Efficient Task-driven Tokens for Continual Knowledge Graph Embedding*. **ECML-PKDD**, 2025.
- [C.12] Yanteng Zhang, Songheng Li, Zeyu Shen, **Qizhen Lan**, Lipei Zhang, Yang Liu, Vince Calhoun. *Lost in Distortion: Uncovering the Domain Gap Between Computer Vision and Brain Imaging—A Study on Pretraining for Age Prediction*. **ISBI**, 2025.
- [C.13] Jong-Im Choi\*, **Qizhen Lan**\*, Qing Tian. *Improving Deep Detector Robustness via Detection-Related Discriminant Maximization and Reorganization*. **WACV**, 2025 (Oral; \* Co-first author).
- [C.14] Zhicheng Ding, **Qizhen Lan**, Qing Tian. *Target-Driven and Student-Centered Knowledge Distillation for Traffic Object Tracking*. **IV**, 2025.
- [C.15] **Qizhen Lan**, Qing Tian. *Gradient-Guided Knowledge Distillation for Object Detectors*. **WACV**, 2024.
- [C.16] Wentao Wang, Xi Xiao, Mingjie Liu, **Qizhen Lan**, Xuanyao Huang, Qing Tian, Swalpa Kumar Roy, Tianyang Wang. *Multi-dimension Transformer with Attention-based Filtering for Medical Image Segmentation*. **ICTAI**, 2024.

## HONORS AND AWARDS

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| • NSF ACCESS Advanced Computing Allocation (≈\$10,000 GPU resources) | 2026 |
| • Graduate College Professional Development Funds                    | 2023 |
| • Analytics Instructional Fee Award                                  | 2018 |

## ACADEMIC SERVICES

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- **Conference Reviewer**

- Annual Conference on Neural Information Processing Systems (NeurIPS), 2026
- ACM International Conference on Multimedia (ACM MM), 2026
- International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2026
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- Association for the Advancement of Artificial Intelligence (AAAI), 2025
- International Symposium on Biomedical Imaging (ISBI), 2025
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024, 2025

- **Journal Reviewer**

- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- Nature - npj Digital Medicine
- IEEE Transactions on Multimedia (TMM)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)